

9. Test Execution

Introduction

Test execution is the process of running the implemented load testing scenarios and collecting performance metrics generated by the system during execution. In this project, Gatling was used to execute the BlazeDemo Flight Booking Application load tests and generate detailed performance reports.

Each test scenario was executed independently using a dedicated Gatling simulation class. During execution, virtual users performed predefined business transactions while Gatling monitored response times, request success rates, throughput, and overall system behavior.

The tests were executed using the following common configuration:

Parameter	Value
Tool	Gatling
Virtual Users	20
Ramp-Up Duration	30 Seconds
Execution Type	Load Testing
Request Validation	HTTP Status Code 200
Report Format	HTML Report

The execution results were automatically generated by Gatling and stored in the target directory for analysis.

9.1 Scenario 1 Execution Report – Search Flights

Objective

The objective of this execution was to evaluate the performance of the flight search functionality under simulated user load.

Workflow Executed

Home Page

↓

Select Departure City

↓

Select Destination City



Find Flights



Display Available Flights

Execution Process

The simulation started by opening the BlazeDemo home page.

Users selected:

- From City: Paris
- To City: Buenos Aires

The flight search request was then submitted using the Find Flights button.

The application processed the request and displayed available flights matching the selected route.

Execution Configuration

Parameter	Value
Virtual Users	20
Ramp-Up Duration	30 Seconds
Requests Executed	Home Page + Find Flights
Validation	HTTP Status 200

Execution Outcome

The scenario executed successfully and all virtual users completed the flight search process without errors.

The generated Gatling report confirmed:

- Successful request execution
- Stable application response
- No transaction failures
- Successful display of available flights

The scenario demonstrated that the application could efficiently process flight search requests under the configured load conditions.
